# 1. Introduction to Computer Network

## Computer Network and its Uses

### *Computer Network*

- Computer network is a large number of separate computers that are interconnected to exchange data and information.
- In computer network, users directly interact with the actual machine to invoke the data exchange and the system do not attempt to make the computers or machines to act coherently.

---

### *Uses of computer network*

1. Business Applications
Resource sharing (Availability of programs, equipments and data to anyone on the network).
E-commerce

2. Home Applications
Access to remote information
Communication
Entertainment

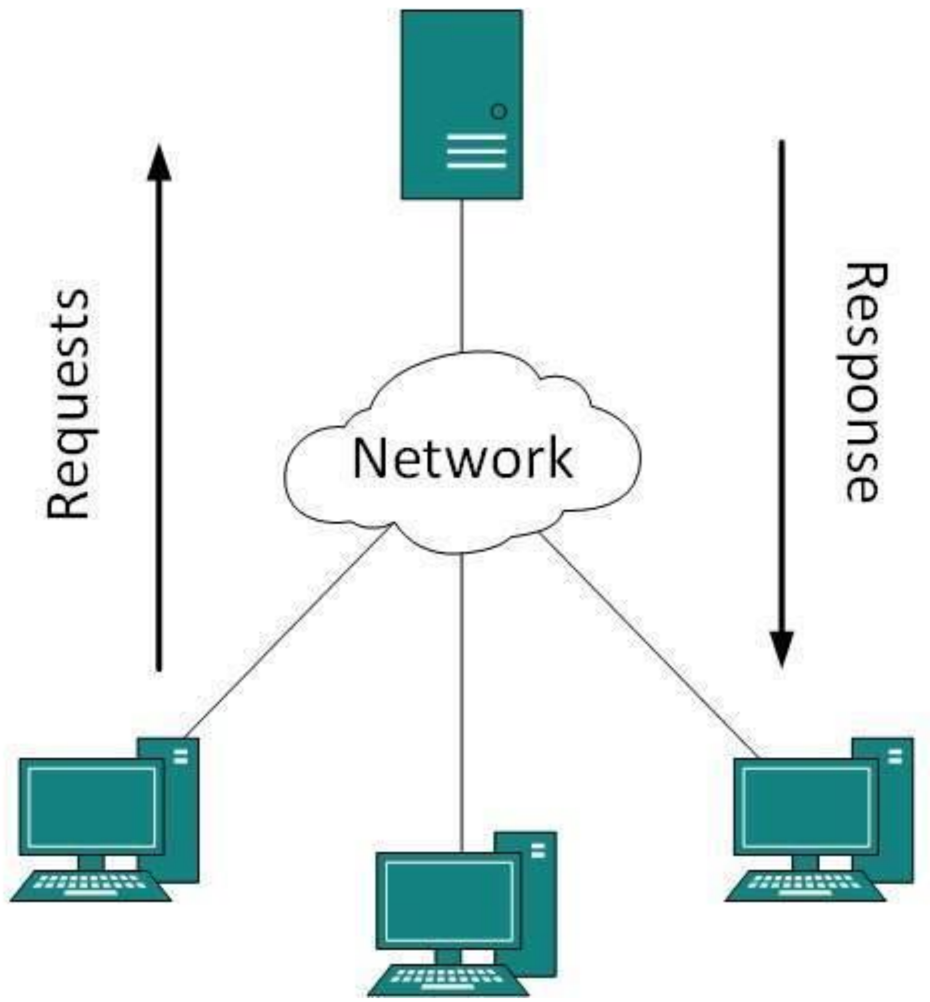
---

## Networking Models

### *Networking Models - Client-Server Model*

- Data are stored in powerful computers known as server.
- Server is managed by a system administrator.
- Users have simple computers called clients.
- Client and server computers are connected by a network.
- Communication occurs as follows:-
1. Client sends request to server over the network.
2. Client waits for a reply.

3. Server gets the request, performs the requested tasks and sends back a reply.
- It is a centralized communication model.
- A single server can have multiple clients associated to it.

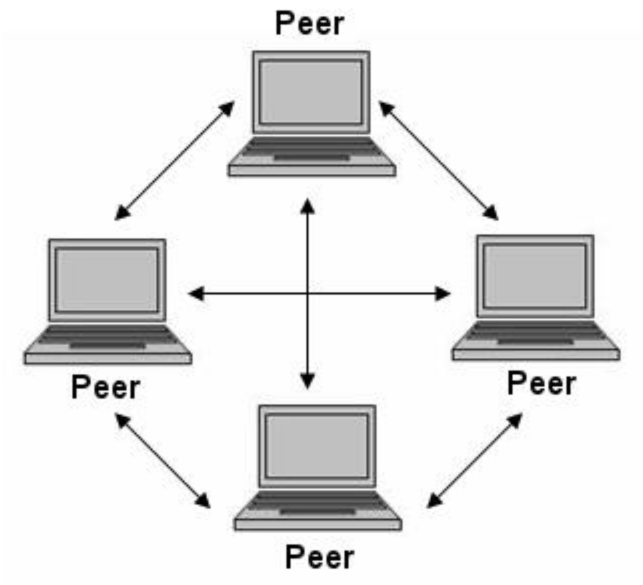



---

***Networking Models - Peer-to-peer Model***

- Peer-to-peer model is a decentralized communication model.
- Every member of a network have equal capacity. It means there is no any distinction of client and server computers.
- For communication, any member is able to initiate the session.
- It is difficult to filter network traffic that which files are being shared.
- It is impossible to administrate the network.

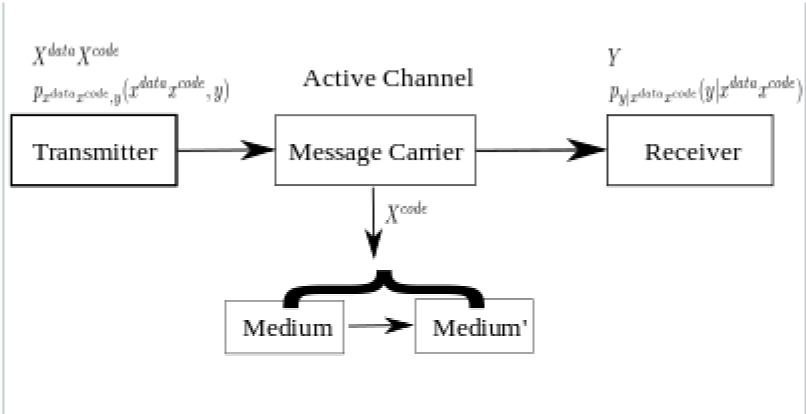
- A failure in the single peer does not fails the whole system.
- It is very much scalable and robust.




---

***Networking Models - Active Network Model***

- Active network model is a communication model in which packets flowing through a network can - dynamically change or modify the operation of the network.
- The packets used are known as active packets.
- Active networking places computation within packets traveling through the network.
- It allows sending code along with packets of information allowing the data to change its form (code) to match the channel characteristics.
- One of the challenges of active networking has been the inability of information theory to mathematically model the active network paradigm and enable active network engineering.




An active network channel uses executable code in the packet to impact the channel controlling the relationship between the transmitted sequence $X$ and the received sequence $Y$. $X$ is composed of a data portion $X^{data}$ and a code portion $X^{code}$. Upon incorporation of $X^{code}$, the channel medium may change its operational state and capabilities.[4]

---

## Protocols and Standards

- Every network is built as a stack of layers, each layer offering some services to the layer above it but do not show the actual implementation of those services.
- When layer n of one computer communicates with layer n of another computer within a network, the set of rules specified for this communication is known as layer n protocol.
- Protocol is a special set of rules that should be followed to communicate within a network.
- When some layer n communicates with layer n of another machine, actual data is not transferred between these layers directly. Such communication is called virtual communication.
- Each layer passes data and information to the layer below it until it reaches lowest layer (physical layer). The actual communication occurs in physical layer via a physically connected mediums.
- Between each pair of adjacent layers, there is an interface that defines the set of services offered by the lower layer to the upper layer.

---

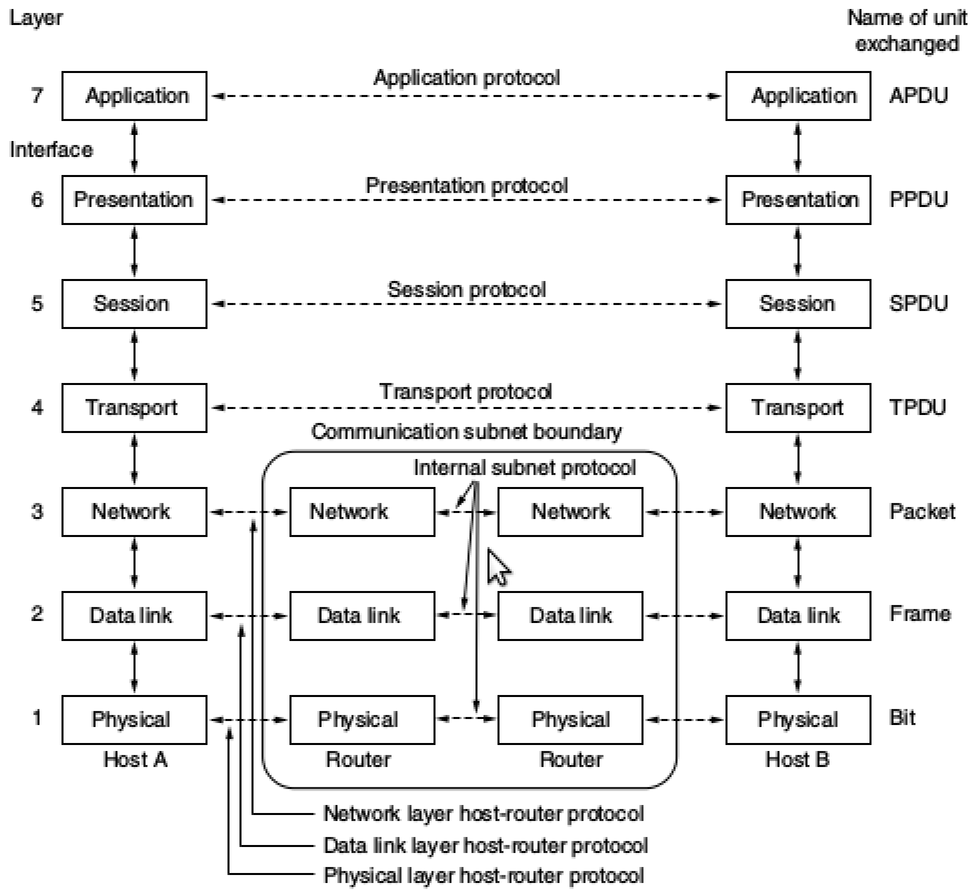
# Comparison of OSI and TCP/IP Model

## *OSI Reference Model*

- The model is called OSI (Open Systems Interconnection) Reference Model because it deals with connecting systems which are open for connection with other systems.
- The OSI model has seven layers: physical, data link, network, transport, session, presentation and application layers.

- The principles involved in formation of these layers are:-
1. A layer should be created where different abstraction is needed.
2. Each layer should perform a well defined functions.
3. The function of each layer should be chosen following the international standard.
4. The layer boundary should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be chosen such that distinct function should be in distinct layers.

- OSI layer does not specify the exact services and protocols to be used by each layer but only defines what each layer must performs.




---

### *TCP/IP Reference Model*

- It consists of four layers. They are link, internet, transport and application layers.
- It provides a packet-switching network based on connectionless layer that runs across different networks.

---

### *Comparison of OSI and TCP/IP Model*

**OSI Model**

- It is a generic, protocol independent standard, acting as a communication gateway between network and end user.
- Transport layer guarantees the delivery of packets.
- It follows vertical approach.
- It has separate presentation and session layers.

- Network layer provides connection oriented and connectionless services.
- Protocols are hidden and are easily replaced as technology changes.
- The services, interfaces and protocols defined are clearly separated.
- It has seven layers.

**TCP/IP Model**
- It is based on standard protocol that allows connection of hosts over the network.
- Transport layer does not guarantee packet delivery.
- It follows horizontal approach.
- It does not have separate presentation and session layers.
- Network layer provides connectionless services.
- Protocol replacing is difficult.
- The services, interfaces and protocols defined are not clearly separated.
- It has four layers.

---

## Example Network - X.25, Frame Relay

### *X.25*

- X.25 is a standard protocol for packet switched WAN communication.
- It contains packet switching exchange (PSE) nodes as networking hardware and plain old telephone service connection or ISDN connection as physical links.
- It contains physical, data link and packet layer.
- It is based on traditional telephony concept of establishing reliable circuits through a shared network.

---

### *Frame Relay*

- Frame relay is a packet switching telecommunication service designed for cost-efficient data transmission for intermittent traffic between LANs and between endpoints in WAN.

- It puts data in variable sized unit called frame along with necessary error correction.
- It provides a permanent virtual circuit as well as switched virtual circuits.
- DTE and DCE are used to transmit data.
- DTE are located at user premise.
- DCE are managed by carriers and provide switching service.
- It is based on X.25 technology.
- It transmits packets at data link layer.

---

***Q) Though we have MAC address, why do we use IP address to represent host in network?***

- MAC address is a low level component of LAN which allow a device to communicate with a machine on the LAN but can not be used to route across the Internet.
IP address is a network layer address which can be used by the routers to figure out where to send data even if it needs multiple hops to reach the destination.

- MAC address works at layer 2 and permits two machines to send messages that are physically connected to each other.

---

***Q) Write the OSI Layer in which following tasks are done:***

Error detection and correction = Transport layer
Encryption and decryption of data = Presentation layer
Logical identification of computer = Network layer
Physical identification of computer = Data link layer
Point-to-point socket connection = Transport layer
Dialogue control = Session layer
Data framing = Data link layer
Timing and voltage of received signals = Physical layer

# 2. Physical Layer

## Physical Layer and its Functions

- Physical layer is a layer of OSI network model which is used to connect one sender with possibly more receivers.
- It is the actual transmission medium of bits.
- It is the only layer that deals with the physical connectivity of two different nodes.
- It defines hardware equipment, cabling, wiring, frequencies and pulses used to represent binary signals.
- The data link layer provides data frames to physical layer which converts them to electrical pulses that represent binary data.

---

***Transmission Impairments***

1. Attenuation
- It is the process of losing strength of signal as it covers distance.
- The receiver needs strong signal in other to interpret the data accurately.

2. Dispersion
- It is the process of spreading and overlapping signals when they travel through a medium.

3. Distortion / Delay
- The bits transmitted through a medium should reach the destination in proper sequence.
- Due to speed and frequency of signal not matched properly, the signal reaches destination in arbitary sequence causing errors in data. This process is called distortion.

4. Noise
- The random disturbance or fluctuation in signal which may distort the actual information is called noise.
- The types of noise are:
a) Thermal noise = Due to heat generated in conductors of a medium.
b) Intermodulation noise = Due to interference caused by multiple

frequencies sharing the same medium.
c) Crosstalk = Due to foreign signal entering into the media.
d) Impulse noise = Due to irregular disturbances like lightening.

---

### *Latency and Throughput*

- Latency is the time taken by a particular packet to reach the destination from source.
- Jitter is delay when the packet has different time every next data transfer.
- Throughput is the amount of data that can traverse through a given medium at a time.
- Throughput is measured in bits per second (bps).

---

### *Digital Transmission*

- Data may be analog or digital.
- For data transmission using computer, it must be converted to digital signals.

---

### *Digital Data to Digital Signal*

1) Line Coding
- It is the process of converting digital data into digital signals.
- Unipolar encoding uses a single voltage level to represent data. To represent 1, high voltage is transmitted and to represent 0, no voltage is transmitted.
- Polar encoding uses multiple voltage levels to represent binary data.
- Bipolar encoding uses three voltage levels ( positive, negative and zero ).

2) Block Coding
- The redundant bits are used to ensure accuracy of received data frames.
- This process is called block coding.

- It is represented by mB/nB ( m bit block is represented with n bit block where n > m.)
- It involves division, substitution and combination.

---

### *Analog Data to Digital Signal*

1) Pulse Code Modulation (PCM)
- It involves three steps:
a) Sampling
b) Quantization
c) Encoding

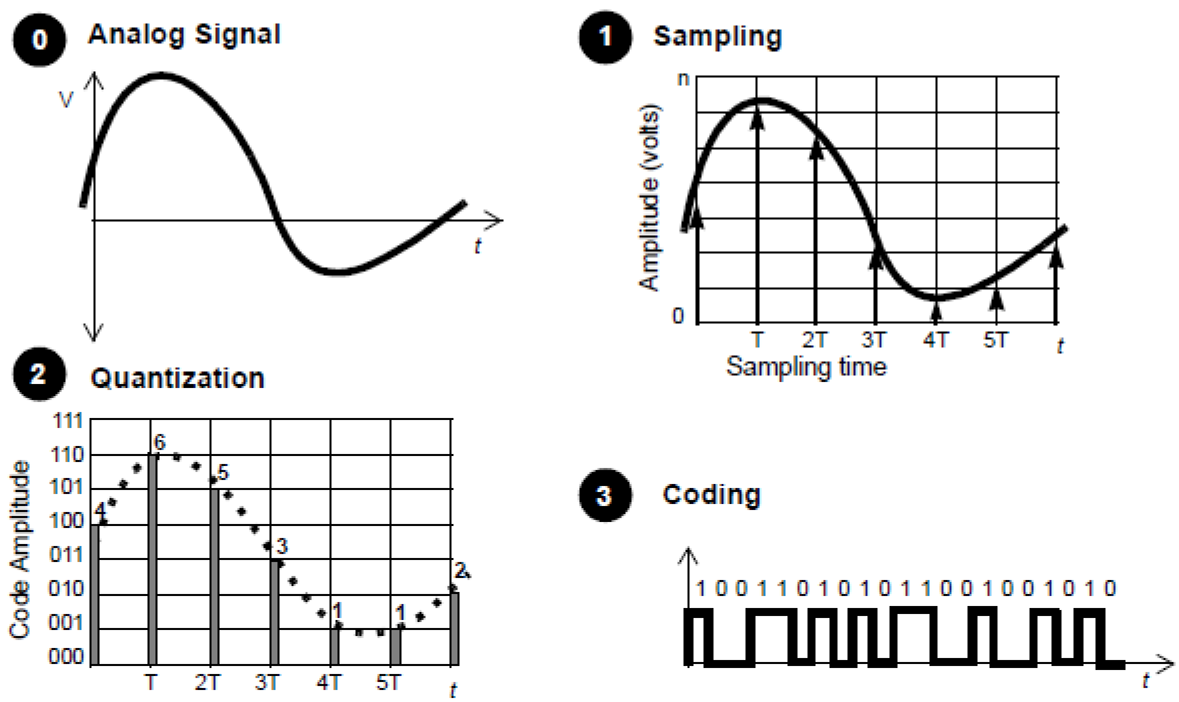



**Figure 1.2** The three steps of digitalization of a signal: sampling of the signal, quantization of the amplitude, and binary encoding.

---

# Transmission Media

- Transmission media is the means through which communication takes place in computer network.
- It is a media over which information is transmitted between two computer systems.
- It is categorized as guided and unguided media.

---

***Twisted Pair Cable***

- It is a transmission media made of two plastic insulated copper wires twisted together to form a single media.
- Only one wire carries actual signal and the other one is used for ground reference.
- The twisting helps in reducing electromagnetic interference and crosstalk.
- It is of two types : STP and UTP cable.
- In computer network, Cat-5, Cat-5e and Cat-6 UTP cables are used, for which RJ45 connectors are used.

---

***Coaxial Cable***

- It is made of two wires of copper.
- The core lies in the center and is made of solid conductor.
- The core is enclosed in an insulating sheath.
- The second wire is wrapped around over the sheath which is also encased by insulator sheath.
- The setup is again covered by plastic cover.
- It is capable of carrying high frequency signals.
- The wrapped structure prevents from noise and crosstalk signals.
- BNC connector is used.

---

***Fiber Optics Cable***

- It works on the principles of total internal reflection of light.
- The core is made of high quality glass.

- From one end light is emitted, it travels through it and at the other end, the light detector detects light stream and converts it to electric data.
- It provides high transmission speed.
- It may be single mode or multimode fibre.
- Single mode is capable of carrying a single ray of light and multi mode can carry multiple beams of light.
- The connectors used are Subscriber channel and Straight tip.

---

### *Line of Sight*

- It is a wireless signal propagation characteristics in which waves travel in a direct path from the source to the receiver.
- It is possible at frequency above 30 MHz.
- Any obstructions between transmitter and receiver will block the signal.

---

### *Satellite*

- Satellite communication uses artificial satellite that relays and amplifies radio telecommunication via a transponder; thus creating a communication channel between transmitter and receiver.
- Wireless communication uses em wares to carry signals which require line of sight and are obstructed by curvature of the Earth.
- The frequency bands are allocated to organizations to minimize the risk of signal interference.

---

## Multiplexing and Switching

### *Multiplexing*

- It is the technique of dividing high capacity medium into low capacity logical mediums so as to process different transmission streams simultaneously over a shared link.

- All the communication mediums are capable of multiplexing.
- Multiplexer is used to divide a physical channel and allocate each channel to each sender.
- Demultiplexer is used to identify each of the data contained in a single medium and sends to different receivers.

---

***Frequency Division Multiplexing***

- It divides carrier bandwidth in logical channels and allocates one user to each channel.
- Each channel has some specific frequency which do not overlap with other channels.
- Guard bands separate the channels.

---

***Time Division Multiplexing***

- The channel is divided into time slots and each user can transmit data within the provided time slot only.
- Both ends must be synchronized.
- Signals from different channels traveled in interleaved path.

---

***Wavelength Division Multiplexing***

- It is conceptually similar to FDM but uses light as signals and wavelength for channel division.
- On each wavelength, TDM can be done.

---

***Code Division Multiplexing***

- It allows users to full bandwidth and transmit signals all the time using a unique code.
- Multiple signals can be transmitted over a single frequency.

---

### *Switching*

- It is the process of forwarding packets coming in from one port to a port leading to destination.
- When data comes on a port, it is called ingress.
- When data leaves a port, it is called egress.
- Connectionless switching is the one in which data is forwarded based on forwarding tables without necessity of handshake.
- Connection oriented switching is the one which establish circuit along path between both ends before forwarding the data.

---

### *Circuit Switching*

- When two nodes communicate with each other over a dedicated communication path, it is called circuit switching.
- It requires a pre-specified route for data transfer and no other data is permitted through that path.
- Circuits must be established for data transfer; which may be temporary or permanent.
- The phases in circuit switching are as follows:
a) Circuit establishment
b) Data transfer
c) Circuit breakdown

---

### *Packed Switching*

- The message is broken down into packets and header of each packet contains switching information.
- The packets can be transmitted independently.
- It does not require more resources to store packets by intermediate devices.

---

### *VC Switching*

- It is a packet switching in which a path is established between source and final destination through which all the packets will be routed.
- For user, the connection appears to be a dedicated physical circuit.
- Other communications can also share parts of the same path.

---

***Q) Difference between circuit switching and packet switching.***

- In packet switching, data is transferred by dividing data into individual packets passing through circuits.
- The route is not exclusively determined.
- Routing algorithms are used.

- In circuit switching, data is transferred through a single dedicated path.
- The route is exclusively determined.
- It requires time to set up channel initially.
- It has predictable and faster performance.
- It has a single point of failure.

---

***Q) Compare switching and multiplexing.***

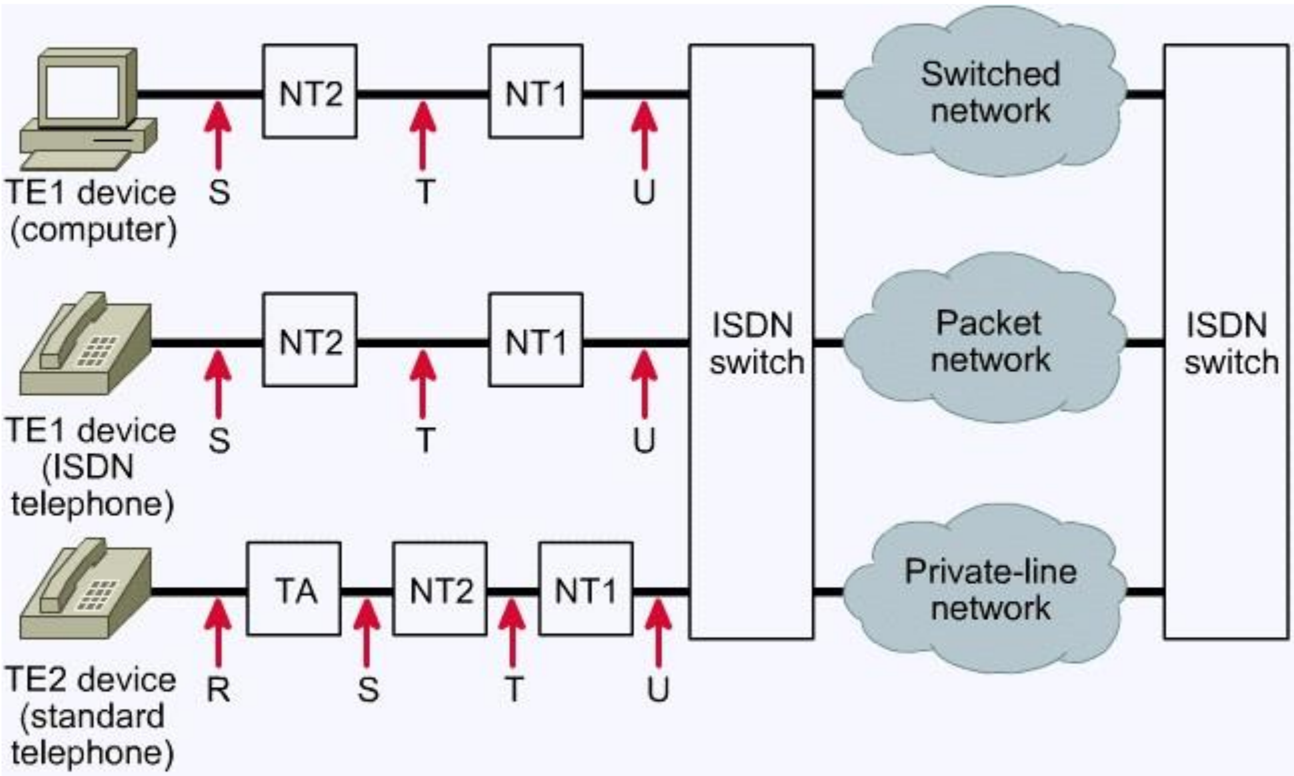
- Multiplexing takes all the inputs and combine them to one output.
- Mux and ‘Demux are used.
- It provides mechanism to share medium among users.

- Switching takes one input and give one output.
- Switching nodes are used.
- It provides mechanism to move data from node to node until they reach destination.

---

# ISDN - Architecture, Interface and Signalling

- ISDN is a digital network which allows digital signals to be sent over existing telephone line.
- It is a communication standards for simultaneous digital transmission of voice, video and data over the traditional circuits of PSTN.
- It is a circuit switched telephone network, which also provides access to packet switched network.




---

### *Components of ISDN*

1) TE1 devices – The devices compatible with ISDN network.
2) TR2 devices – The devices incompatible with ISDN network.
3) TA – Terminal Adapter that converts signal making it ISDN compatible.
4) NT1 – Connects 4-wire ISDN to 2-wire local loop.
5) NT2 – Directs traffic to and from subscriber devices and NT1.

---

### *Interfaces of ISDN*

1) R – Connection between TE2 device and TA.
2) S – Connection between devices and NT2.
3) T – Outbound connection from NT2 to NT1.
4) U – Connection between NT1 and ISDN network owned by the telephone company.

# 3. Data Link Layer

## Data Link Layer and its Functions

- It is the second OSI layer that is responsible to hide the underlying hardware details and acts as a medium to communicate for upper layers.
- On sending end, it converts data stream to signals bit by bit and sends over the underlying hardware.
- On receiving end, it picks up data from hardware in the form of signals, assembles them in a recognizable frame format and hands over to upper layer.
- It has two sub layers – logical link control and media access control.

---

***Functions of Data Link Layer***

Framing
- It receives packets from network layer and encapsulates them into frames, which is send on hardware bit by bit.
- On receiver, it gets signals from hardware and assembles them into frames.

Addressing
- It provides hardware addressing mechanism which is unique and encoded into hardware at the time of manufacturing.

Synchronization
- It synchronizes both computers when sending data frames on the link for successful transfer.

Error Control
- It helps in detecting errors and recovering actual data bits.
- It provides error reporting mechanism to the sender.

Multi Access
- It provides mechanism for accessing shared media among multiple systems reducing data collision.

Flow Control
- It ensures flow control to enable both devices to exchange data on same speed.

---

## Framing

- It deals with breaking of bit stream up into frames.
- The general structure of data link layer frame is shown below:

```
    | MAC Header | LLC Header | Layer 3 protocol data |
MAC Trailer |
```

---

***Approaches for framing***

1. Character Count
- It uses a field in the header to count the number of characters in the frame.
- On receiver end, the layer sees the character count and knows how many characters forms a frame.
- Count can be changed due to transmission error that results in receiver to be out of synchronization.

2. Flag bytes with byte stuffing
- It provides a special byte called FLAG at the beginning and the end of each frame.
- Two consecutive flag bytes indicate the end of one frame and start of next frame.
- If error is detected, it looks for flag byte to determine end of current frame.
- The general frame structure is :

```
    | FLAG | HEADER | PAYLOAD FIELD | TRAILER | FLAG |
```

- When binary data, object program or numbers are transmitted, the flag byte pattern may occur in the data.
- It can be resolved by adding special escape byte (ESC) by the sender before every accidental flag byte in the data.
- The data link layer of receiver removes this ESC byter before providing data to network layer. This process is called byte stuffing.
3. Flag with bit stuffing
- Each frame begins and ends with a special bit pattern called flag byte. Eg: 01111110
- If sender's data link layer encounters five consecutive 1's in the data, it automatically stuffs a 0 bit into outgoing stream.
- If receiver sees 0 bit following five consecutive 1 bits, it automatically removes 0 bit.
- This process is called bit stuffing.

```
     Original Data :  0110 11111 11111 11111 10011
     Data on the line:     0110 111110 111110 111110 10
011
     Data stored by receiver:    0110 11111 11111 11111
10011
```

---

## Error Detection and Correction

- Error detection is used by the receiver to determine if a packet contains error.
- Error correction is used by the receiver to correct the errors found in the packet. It involves retransmission requests to the users.

---

### *Types of error*

- Single bit error is an error in which only one bit in a frame is corrupted.

- Multiple bits error is an error in which more than one bit is corrupted in a frame.
- Burst error is an error containing more than one consecutive corrupted bits in a frame.

---

***Error Detection Techniques:***

1) Parity Check
- One extra bit (even parity or odd parity bit) is sent along with original bits to make number of 1’s either even or odd.
1 0 0 1 0 0 1 = 1 0 0 1 0 0 1 1 (Even Parity)
- On receiver, if count of 1’s is odd in above case, the frame is considered to be corrupted.
- If more than one bits is corrupted, it is hard to detect the error.

2) Cyclic Redundancy Check (CRC)
- It involves binary division of the data bits being sent.
- The divisor is generated using polynomials.
- Sender performs division and calculates remainder which is added at the end of actual bits.
- The actual data plus the remainder is called codeword.
- The receiver conducts division on codeword and if remainder is all 0, it is accepted.

---

***Error Correction Techniques:***

- Backward Error Correction (Request sender for data retransmission on error detection.)
- Forward Error Correction (Execute error correcting code for auto recovery.)

---

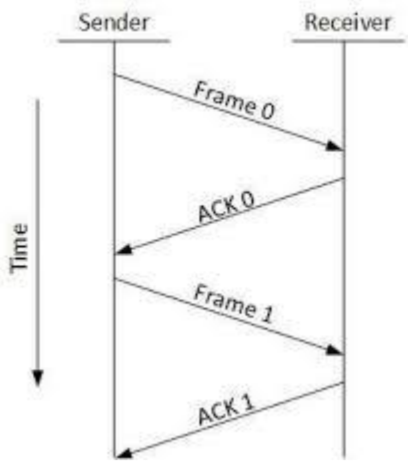
# Flow Control

- It ensures that sender sends data at a speed on which the receiver can process and accept the data.

---

***Mechanisms for flow control: Stop and Wait***

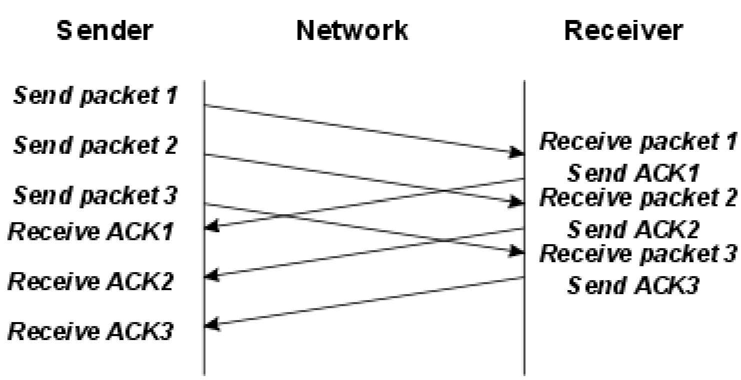
- It forces the sender to stop and wait after transmission of data frame until the ack is received.




---

***Mechanisms for flow control: Sliding Window***

- Both sender and receiver agree on number of data frames after which ack should be sent.




---

***Mechanisms for error control***

1) Stop and Wait ARQ
2) Go-Back-N ARQ
3) Selective Repeat ARQ

---

## HDLC

- It is a bit oriented code transparent synchronous data link layer protocol.
- It provides both connection-oriented and connectionless services.
- It is used for point-to-multipoint connections.
- The structure of HDLC frame is:

```
     | Flag(8) | Address (8+) | Control(8/16) | Informat
ion(n*8) | FCS(16/32) | Flag (8) |
```

-HDLC frames can be information, supervisory or unnumbered frames.
- The types of link configurations are:
a) Normal Response Mode
b) Asynchronous Response Mode
c) Asynchronous Balanced Mode

---

## PPP

- It is a data link protocol used to establish a direct connection between two nodes.
- It provides connection authentication, transmission, encryption and compression.
- It is used over Internet access connections.
- The requirement for PPP is that the circuit provided must be duplex.
- The structure of PPP frame is:

```
| Flag(8) | Address(8) | Control(8) | Protocol(16) | Inf
ormation(n*8) | Padding(n*8) | FCS(16/32) | Flag (8) |
```

---

## Data Link Layers - Sub Layers

***Media Access Control:***

- It is responsible to share physical connection to the network among several computers.
- It helps in frame delimiting and recognition.
- It provides protection against errors.
- It addresses destination stations.

---

***Logical Link Control:***

- It provides node to node flow control and error control.
- It provides multiplexing protocols.
- It manages traffic over physical medium.

---

## Channel Allocation Problem

- Channel allocation means which station is permitted the channel next.
- It may be:
a) Static (assignment for long duration)
b) Dynamic (stations continuously compete for allocation)

- In static allocation, channel is requested, data is transferred and channel is released.
- Channel becomes private after allocation.
- Channel can be divided using FDM and TDM.

---

## Multiple Access Protocol

- It is a protocol that coordinate access to the link, which can be used by both wired and wireless network.
- Nodes use multiple access protocols to regulate transmission onto the shared broadcast channel.
- All frames involved in the collision are lost.

---

## Ethernet

- It is a working group and a collection of IEEE standards defining physical layer and MAC sub layer of wired Ethernet.
- It defines LAN access method using CSMA/CD.
- Ethernet is a link layer protocol describing how networked devices on same network can format data for transmission and how to put that data on the network connection.

---

## FDDI; ALOHA; VLAN; CSMA/CD; Token Bus; Token Ring and Wireless LAN

### *FDDI*

- FDDI stands for Fiber Distributed Data Interface.
- It is a set of ANSI and ISO standards for data transmission on fiber optic in LAN.
- FDDI protocol is based on token ring protocol.
- It consists of two token rings.
- Primary ring offers up to 100 Mbps capacity.
- Secondary ring is for backup if primary ring fails.
- It offers circuit switched service to the network.

---

### *ALOHA*

- ALOHA stands for Additive Links On-line Hawaii Area.
- It uses random method of medium access and UHF for its operation.
- The following assumptions are made:
a) All the frames have same length.
b) Stations can not generate a frame while transmitting.
c) The population of stations attempts to transmit according to Poisson Distribution.

---

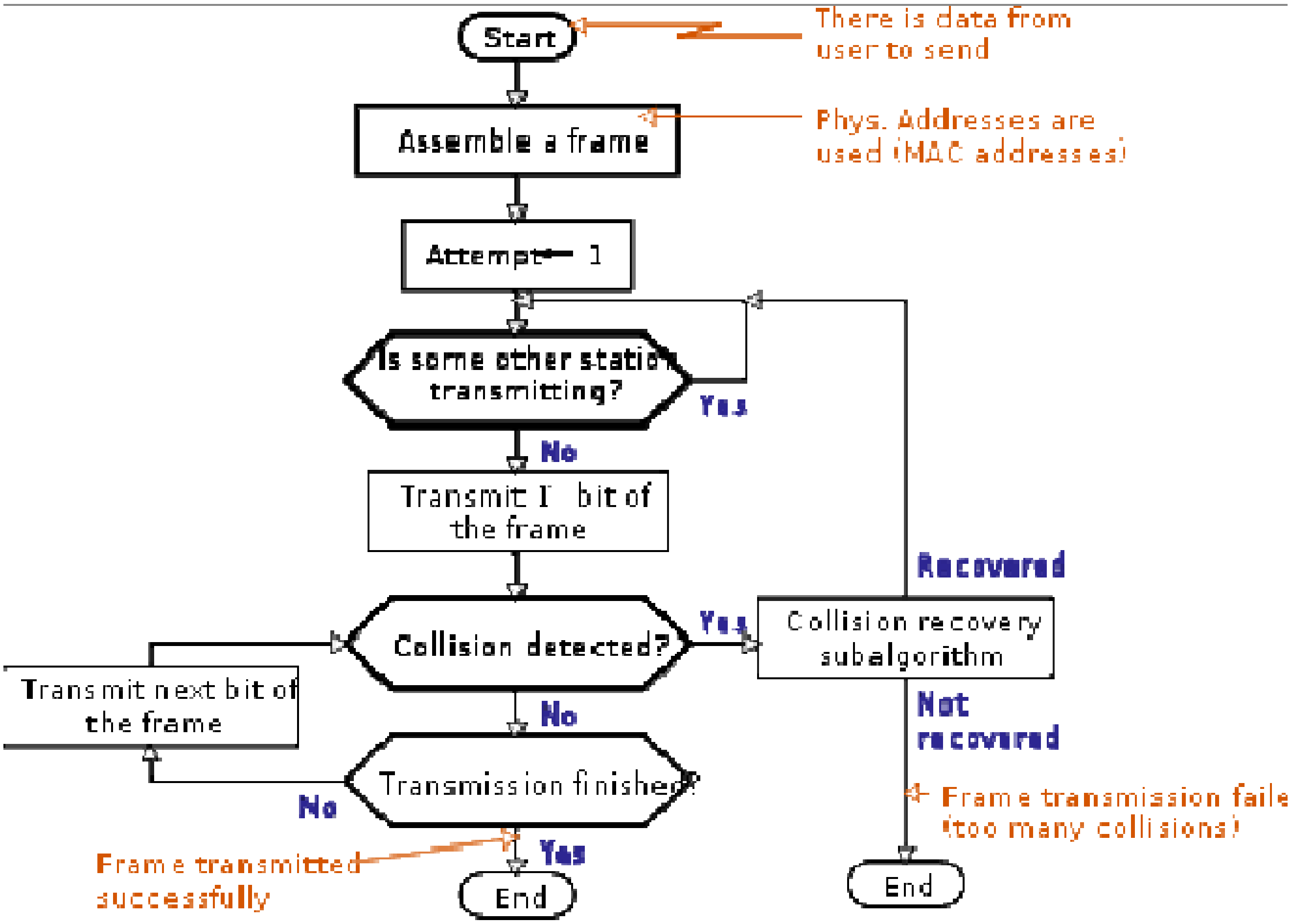
***VLAN***

- VLAN stands for Virtual Local Area Network.
- It is a broadcast domain partitioned and isolated in a computer network.
- Network equipment is configured to subdivided network into VLAN.
- It allows network administrators to group hosts together if the hosts are not on the network switch also.

---

***CSMA/CD***

- CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection.
- It is a media access control method which uses carrier sensing scheme in transmitting station to detect collision.
- If collision is detected, station stops transmitting the frame, transmits the jam signal and waits for random time interval before trying to resend the frame.




### *IEEE 802.4 (Token Bus)*

- Token bus is a network implementing token ring protocol over a virtual ring on a coaxial cable.
- A token is passed around the network nodes and the node with token is only able to transmit.
- If a node does not have anything to send, it passes token to next node on virtual ring.
- Each node must know address of its neighbor in the ring.
- The endpoints of the bus do not meet to form a physical ring.

### *IEEE 802.5 (Token Ring)*

- Token ring is a LAN communication protocol using 3 byte frame called token that travels around a logical ring of nodes.
- It provides fair access for all stations and eliminates the collisions.

Multiple Access in 802.5
- Time Division Multiple Access is used.
- Stations access channel in rounds with the help of tokens.
- Each station gets fixed length slot in each time.
- The unused slots go idle.

---

***IEEE 802.1 (Wireless LAN)***

- Wireless LAN is a LAN that uses radio waves to communicate.
- It is defined in standard IEEE 802.1.

# 4. Network Layer

## Internetworking and devices

### *Network Layer*

- Network layer is the layer 3 of OSI model which manages options pertaining to host and network addressing, managing sub networks and inter-networking.
- It is responsible for routing packets from source to destination within or outside a subnet.

---

### *Functions*

- Addressing devices and networks.
- Populating routing tables.
- Queuing incoming and outgoing data.
- Inter-networking between two different subnets.
- Delivering packets to destination.

---

### *Inter-networking*

- Routing between two networks of same kind or different kinds is called inter-networking.
- Tunneling is a mechanism by which two or more same networks communicate with each other passing through intermediate networking complexities.
- When data enters from one end of tunnel, it is tagged.
- The tagged data is routed inside transit network.
- When data exits the tunnel, the tag is removed and delivered to the other part of the network.

---

### *Networking Devices*

Repeater
- Repeater is an electronic device that receives a signal and retransmits it.
- It helps to regenerate or replicate weak or distorted signals.
- It operates at physical layer.
- It only repeats the signal without understanding the packets.
- It does not have physical address on the network.

Hub
- Hub is a network component that acts as a common connection point for nodes in a network.
- It contains multiple ports.
- When a packet arrives at one port, it is copied to all other ports so that all segments of LAN can see all packets.

Switch
- When a packet arrives at one port, it is copied to only the port that leads to destination node.

Bridge
- A network bridge is a network component that creates a single aggregate network from multiple networks.
- It operates in physical and data link layer.
- It operates using physical address of machines.
- The bridge creates function table with machine’s address and the segment they belong to.

Router
- Router is a network device that forwards data packets between networks.
- A router is connected to two or more data lines from different networks.
- When a data packet comes in on one of the lines, it reads the address information in the packet to determine the ultimate destination.
- The information in routing table helps to direct packet to next network.

Gateway
- Gateway is an inter-networking system capable of joining together two networks that use different base protocols.

---

# Internet Address and Classful Address

### *Internet Addressing*

- Network address is logical address given by the software which can be changed by appropriate configurations.
- A network address always points to host/node/server.
- It is configured on network interface card.
- It is mapped by system with MAC address of machine.
- Eg: IP addresses

- IP address provides mechanism to differentiate between hosts and network.
- As IP address is assigned in hierarchial manner, a host always resides under a specific network.

- A host in different subnet need a mechanism to locate each other, which is done by DNS.
- DNS is a server which provides layer 3 address of remote host mapped with its domain name.
- When a host acquires layer 3 address of remote host, it forwards packets to its gateway.

---

### *Classful Addressing*

- The IP address is 32 bits in size.
- There are five classes:

```
Class       First Octet
A           0-127
B           128-191
```

```
C            192-223
D            224-239
E            240-255
```

- Class D and E are not used generally.
- The class for a new network is assigned based on the size of the network.

---

## Subnetting

- Subnetting is the process of breaking the class of network into smaller network/subnet.
- It is necessary because a large single network of a class without subnet will not be efficient.
- It splits the host field into subnet creating a three part address.
- The network field remains unchanged which is determined by classful addressing.
- A subnet mask is assigned to determine between subnet and host fields.

```
Class        Subnet Mask
A            255.0.0.0 or /8
B            255.255.0.0 or /16
C            255.255.255.0 or /24
```

- In subnet mask, 1 represents network bit and 0 represents host bit.

---

***Q) Allocate 30, 24, 25 and 12 IP addresses to four department with minimum wastage. Specify range of IP address, broadcast address, network address and subnet mask for each department from address pool 202.77.19.0/24.***

```
The starting IP address is : 202.77.19.0/24
The network is of class C.
The subnet mask is 255.255.255.0 (i.e. /24)


Since there are four departments, the network design req
uires 4 subnets.


Using Variable Length Subnet Mask (VLSM),


For Department A;
To support 30 hosts, it will require 32 IP address such
that:
2^y = 32 => y = 5
So, we need 5 bits for host field. Hence it requires /27
mask.


For Department B;
To support 24 hosts, it will require at least 26 IP addr
ess such that:
2^y = 32 => y = 5
So, we need 5 bits for host field. Hence it requires /27
mask.


For Department C;
To support 25 hosts, it will require at least 27 IP addr
ess such that:
2^y = 32 => y = 5
```

```
So, we need 5 bits for host field. Hence it requires /27
mask.



For Department D;
To support 12 hosts, it will require at least 14 IP addr
ess such that:
2^y = 16 => y = 4
So, we need 4 bits for host field. Hence it requires /28
mask.



Dpt    IP AddressNetwork Address  Range of IP for host
Broadcast Address          Subnet Mask
A     202.77.19.0202.77.19.0       202.77.19.1-202.77.19
.30     202.77.19.31     255.255.255.224
B     202.77.19.32   202.77.19.32        202.77.19.33-202
.77.19.62     202.77.19.63     255.255.255.224
C     202.77.19.64   202.77.19.64        202.77.19.65-202
.77.19.94     202.77.19.95     255.255.255.224
D     202.77.19.96   202.77.19.96        202.77.19.97-202
.77.19.111     202.77.19.112       255.255.255.240
```

---

## Static and Dynamic Routing; Routing Table

### *Routing*

- Whenever a device has multiple paths to reach the destination, it always selects one path. This process is called routing.
- It is done by router.
- A router is configured with some default route.
- The default route forwards a packet if no route is found for specific destination.

- If multiple paths exists to reach same destination, decision is based on : hop count, bandwidth, metric, prefix length and delay.

---

### *Static and Dynamic Routing*

- Static routing is when you statically configure a router to send traffic to particular destinations in predetermined directions.
- It provides default route.

- Dynamic routing is when you use a routing protocol to figure out the best possible route for the traffic.
- It can provide the best route.

---

### *Routing Table*

- Routing table is a set of rules in a table format which is used to determine where data packets traveling over an IP network will be directed.
- It contains all the information necessary to forward a packet along the best path toward its destination.
- A basic routing table includes following informations:
a) Destination IP address
b) Next hop IP address
c) Outgoing network interface used
d) Cost metric to each available route
e) Routes
- Routing table can be maintained manually or dynamically.
- Dynamic routing tables allow devices to respond to device failures and network congestion.

---

## Routing Protocol - RIP, OSPF, BGP, Unicast and Multicast Routing Protocol

### *Routing Information Protocol (RIP)*

- RIP is a protocol that defines a way for routers, which connect networks using IP, to share information about how to route traffic among networks.
- Each router maintains a routing table which consist of a list of all destinations it knows how to reach and the distance to that destination.
- It uses distance vector algorithm to decide the route of packet to its destination.
- If it receives update on a route with shorter path, it will update its routing table with length and next hop address of the shorter path.
- If new route has longer path, it waits through a hold-down period and only update the table if the new route is stable.
- It follows a state of convergence. Each router sends its entire routing table to its closest neighbors every 30 seconds until all RIP hosts within the network have same knowledge of routing paths.
- It can know about router crash and if router stops sending update for six successive cycle, it will be dropped from the route.
- It uses modified hop count to determine network distance.

---

### ***Open Shortest Path First (OSPF)***

- It is a protocol for routers connecting networks using IP, used to find the best path for packets as they pass through a set of connected networks.
- A router which detects a change to a routing table immediately multicasts the information to all other OSPF hosts.
- It only sends the part that has changed. This helps to minimize convergence time.
- It accounts router hop as well as other network information like cost metric to determine best path.
- It has RIP support built in for compatibility with older networks using RIP.

---

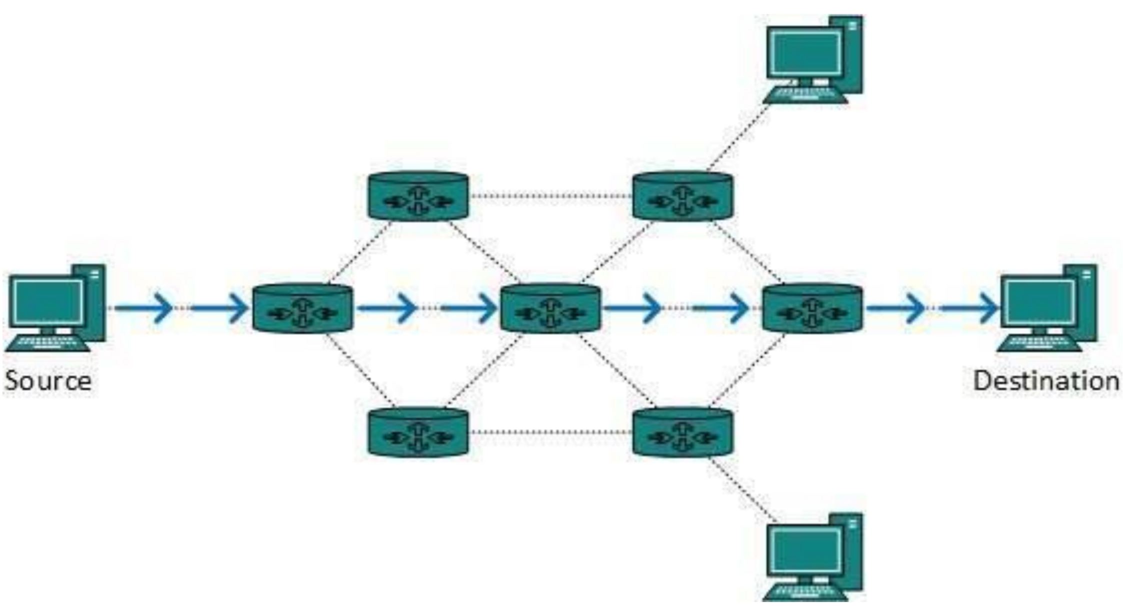
### ***Border Gateway Protocol (BGP)***

- It is a protocol that manages how packets are routed across the Internet through the exchange of routing and reachability information between edge routers.
- It makes routing decision based on path, rules or network policies configured by a network administrator.
- Each BGP router maintains a standard routing table used to direct packets in transit.
- The table is used in conjunction with separate routing table called routing information base (RIB).
- It is based on TCP/IP and uses client-server topology.

---

### *Unicast Routing*

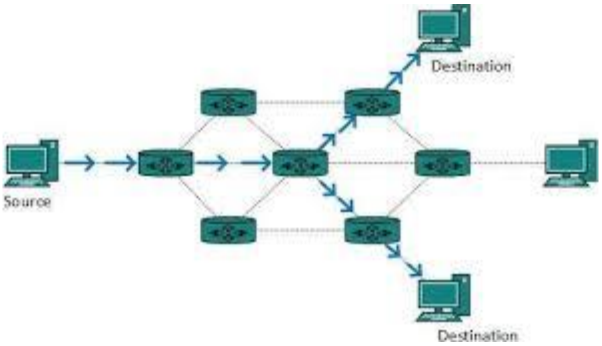
- Unicast routing is the process of routing unicast data (data sent with specified destination) over the Internet.
- The destination is already known.
- The router just has to look up the routing table and forward packet to next hop towards destination.
- Eg: Distance vector routing protocol and link state routing protocol.




---

### *Multicast Routing*

- Multicast routing is a type of broadcast routing in which the data is sent to only nodes which wants to receive the packets.
- It uses spanning tree protocol to avoid looping.
- It also uses reverse path forwarding technique.
- Eg: Multicast OSPF, Core based tree, etc.




---

## Routing Algorithms - Shortest Path, Flooding, Distance vector Routing, Link State Routing

### *Shortest Path Algorithm*

- Shortest path can be calculated only for weighted graphs.
- The edge connecting two vertices can be assigned a non-negative real number called weight of edge.
- The algorithm is:
a) Initialize array smallestWeight so that
smallestWeight[u] = weights[vertex, u]
b) Set smallestWeight[vertex] = 0.
c) Find vertex v, that is closed to vertex for which shortest path has not been determined.
d) Mark v as next vertex for which smallest weight is found.
e) For each vertex w in G, such that shortest path from vertex to w has not been determined and edge (v, w) exists, if weight[u, w] < current weight, update weight of w to weight of v + weight of edge (v, w).

*Example:*

```
                              B
                         /         \
                      1               2
                   /                     \
Source(A) -- ---- 5 --- (D)Destination
                  \                      /
                    2                  1
                      \             /
                             C


Edge         Cost           Path
-------------------------------------------
B                1                A-B
C                2                A-C
D                5                A-D
(Choose path A-B)


Edge         Cost           Path
-------------------------------------------
B                1                A-B
C                2                A-C
D                3                A-B-D
(A-B-D < A-D, So, A-B-D path is selected.)
```

---

***Flooding Algorithm***

- It is the static routing algorithm.
- Every incoming packet is sent on all outgoing lines except the line on which it has arrived.
- It generates a large number of duplicate packets on the network.

---

***Distance Vector Routing***

- Each node constructs a one dimensional array containing distances to all other nodes and distributes that vector to its immediate neighbors.
- The starting assumption is each node knows the cost of the link to directly connected neighbors only.
- A link that is down is assigned an infinite cost.
- Every node sends a message to its direct neighbors containing its personal list of distance.
- If any recipient finds the sender has path shorter than the one they know, they update the new path length.
- It should know which node tell them about the path they use.
- Each node maintains forwarding table.

```
      All path cost is 1.
                     A --------- B
                   / | \              /
                 /   |   \          /
               /     |     \      /
             E     F         C
                     |         |
                     |         |
                     G----D



Info stored at Node             Distance to reach Node
                                                A     B
C     D     E     F      G
------------------------------------------------------------
---------------------------------------
A                                             0       1
1     ?      1     1        ?
B                                             1       0
1     ?      ?     ?        ?
C                                             1       1
```

```
0     1       ?     ?        ?
D                                            ?      ?
1     0       ?      ?        1
E                                            1      ?
?     ?       0      ?        ?
F                                            1      ?
?     ?       ?      0        1
G                                            ?      ?
?     1       ?     1        0


Now, On update:
Info stored at Node            Distance to reach Node
                                               A     B
C     D      E     F      G
-------------------------------------------------------
---------------------------------------
A                                            0      1
1     2       1     1        2
B                                            1      0
1     2       2     2        3
C                                            1      1
0     1       2     2        2
D                                            2      2
1     0       3     2        1
E                                            1      2
2      3       0     2        3
F                                            1      2
2      2       2     0        1
G                                            2      3
2     1       3     1        0


Forwarding table for node B:
```

```
Destination          Cost          Next Hop
-------------------------------------------------------
A                    1             A
C                    1             C
D                    2             C
E                    2             A
F                    2             A
G                    3             A
```

---

***Link State Routing***

- Each router knows about its direct neighbors.
- Each router constructs a link state packet (LSP), which consists of:
a) ID of node that create LSP.
b) A list of direct neighbors and cost of link to each one.
c) Sequence number
d) A time to line (TTL) for the packet.

- Link state flooding is done.
- Each router stores most recently generated LSP from each other router.
- Shortest path route to each destination is computed.

---

***Q) Compare distance vector routing and link state routing.***

- Distance vector routing sends the entire routing table to the directly connected neighbors.
- It has slow convergence.
- It is susceptible to routing loops.
- The updates are sometimes sent using broadcast.
- It does not know the network topology.
- It is simple to configure.
- Eg: RIP

- Link state routing only sends the link state information to the directly connected neighbors.
- It has fast convergence.
- It is less susceptible to routing loops.
- The updates are always sent using multicast.
- It knows the entire network topology.
- It is difficult to configure.
- Eg: OSPF

---

## ARP, RARP, IP and ICMP

### *Address Resolution Protocol (ARP)*

- ARP is a protocol used by IPv4 to map IP addresses to the hardware addresses used by a data link protocol.
- It operates below network layer.
- It is a part of interface between network layer and data link layer.
- An ARP cache table is used to maintain each MAC address and its corresponding IP address.
- When an incoming packet destined for a host on a LAN arrives at a gateway, the gateway asks ARP program to find MAC address that matches the IP address.
- ARP program looks in ARP cache and if it finds the address, it provides address to gateway. If not found in cache, ARP program broadcasts a request packet in a special format to all hosts on the LAN. A machine that owns the IP address returns a reply.
- The gateway then convert the incoming packet to the right format and length; then sent to the machine.

---

### *Reverse ARP (RARP)*

- It is a protocol used by a physical machine in a LAN to request to learn its IP address from the ARP cache.
- When a new machine is set up, its RARP client requests RARP server on router to send its IP address.

---

### *Internet Control Message Protocol (ICMP)*

- It is an error reporting protocol used by network devices to generate error messages to the source IP address when error prevents packet delivery.
- All the IP network device is capable to send, receive and process ICMP messages.
- ICMP messages are transmitted as datagrams which consists of IP header encapsulating ICMP data.

---

### *Internet Protocol (IP)*

- IP is a protocol by which data is sent from one computer to another on the Internet.
- Each computer on the Internet has at least one IP address that uniquely identifies it from all other computers.
- IP is responsible for delivery of data packets without any consideration of its order.
- The order of packets are managed by TCP.
- IP is connectionless protocol.

---

### *Q) What is routed and routing protocol?*

- A routed protocol is a protocol by which data can be routed.
- It should have addressing scheme and subnetting.
- It is used by all the hosts of inter-network.
- Eg: IP, AppleTalk, IPX

- A routing protocol is a protocol that makes the router able to build and maintain routing tables.
- It is used by routers only.
- Eg: Distance vector and link state protocols.

# 5. Transport Layer

## Transport Services

- Transport layer is layer 4 of OSI model which is responsible for peer-to-peer connection between two processes on remote hosts.

---

***Functions of Transport Layer***

- The functions of transport layer are:
a) It breaks the data obtained from application layer into smaller segments.
b) It ensures that data must be received in same sequence in which it was sent.
c) It provides end-to-end delivery of data between hosts.

---

***Services provided by Transport Layer***

- The services provided to upper layers are:
a) Addressing (handles addressing of the processes on the node.)
b) Connection management (establish and release connections.)
c) Flow control and buffering
d) Multiplexing

---

## UDP

- UDP uses minimum amount of communication mechanism.
- The receiver does not generate an ack of packet received.

---

***Requirements of UDP***

- UDP is deployed where ack packets share significant amount of bandwidth along with actual data.

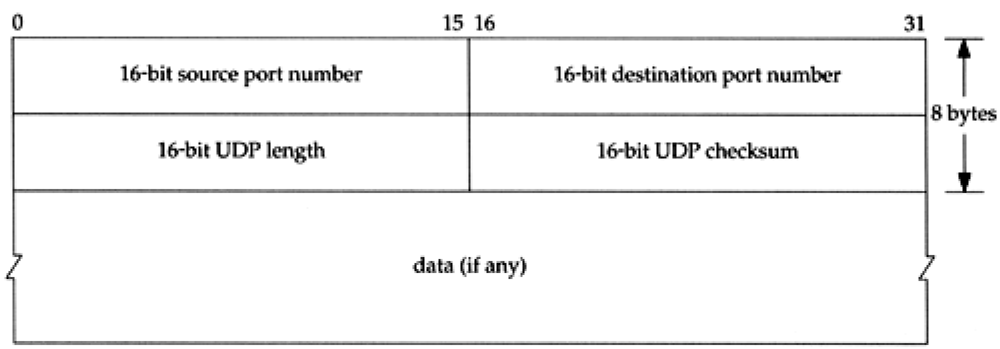
- Even if some packets are lost, it can be ignored easily.
- Eg: in video streaming

---

***Features of UDP***

- It is used if ack of data does not have significance.
- It flows data in one direction.
- It is suitable for query based communication.
- It is connectionless.
- It does not provide congestion control mechanism.
- It does not guarantee data delivery in order.
- It is stateless.

---




***UDP Header***

- Source port identifies the port of application process of sender.
- Destination port identifies the port of application process of receiver.
- Length specify the entire length of UDP packet with header.
- Checksum stores checksum value generated by the sender before sending.

---

## TCP

***Features of TCP***

- TCP is reliable protocol as the receiver always send positive or negative ack about data packet to the sender such that it knows whether data packet reach destination or it needs to resend it.
- It ensures that data reaches intended destination in proper order.
- It requires connection between two remote ends to be established before sending actual data.
- It provides error checking and recovery mechanisms.
- It provides flow control and quality of service.
- It operates in client-server point-to-point mode.
- It provides full duplex server (it can perform roles of both receiver and sender).

---

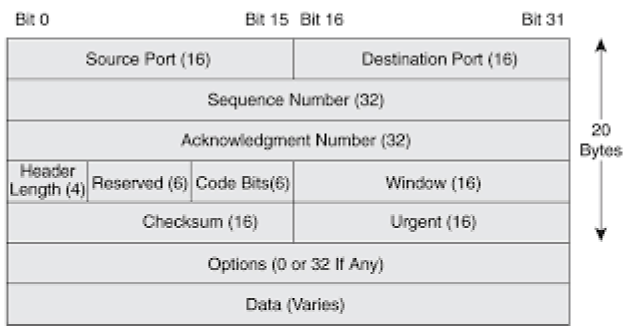
***TCP Header***

- The length of TCP header is minimum of 20 bytes and maximum of 60bytes.
- Source port identifies port of application process of sender.
- Destination port identifies port of application process of receiver.
- Sequence number of data bytes of segment.
- Acknowledge no contains next sequence number of data byte expected when ACK flag is set.
- Data offset implies size of TCP header and offset of data in current packet.
- Reserved is for future use and all are set zero by default.
- Flags

a) None Sum (NS) = Explicit congestion notification signaling process.
b) Congestion Window Reduced (CWR) = It is set when a host receives packet with ECE bit set to acknowledge that ECE is received.
c) ECE = If SYN bit is 0, IP packet has CE bit set. If SYN bit is 1, the device is ECT capable.
d) URG = Urgent pointer field has significant data and should be processed if URG is set.
e) ACK = Acknowledgement field has significance if set.
f) PSH = When set, it is a request to receiving node to push data to application without buffering.
g) RST = It is used to refuse incoming connection, reject a segment

and restart a connection.
h) SYN = It is used to setup connection between hosts.
i) FIN = It is used to release connection.

- Window size is used to control flow and indicates amount of buffer in bytes the receiver has allocated for a segment.
- Checksum contains checksum of header (data and pseudo header).
- Urgent pointer points to urgent data byte if URG flag is set.
- Options provide additional options.




---

### *Addressing in TCP*

- TCP communication between two remote hosts is done by means of port numbers known as Transport Service Access Points.
- Port numbers ranges from 0 to 65535.

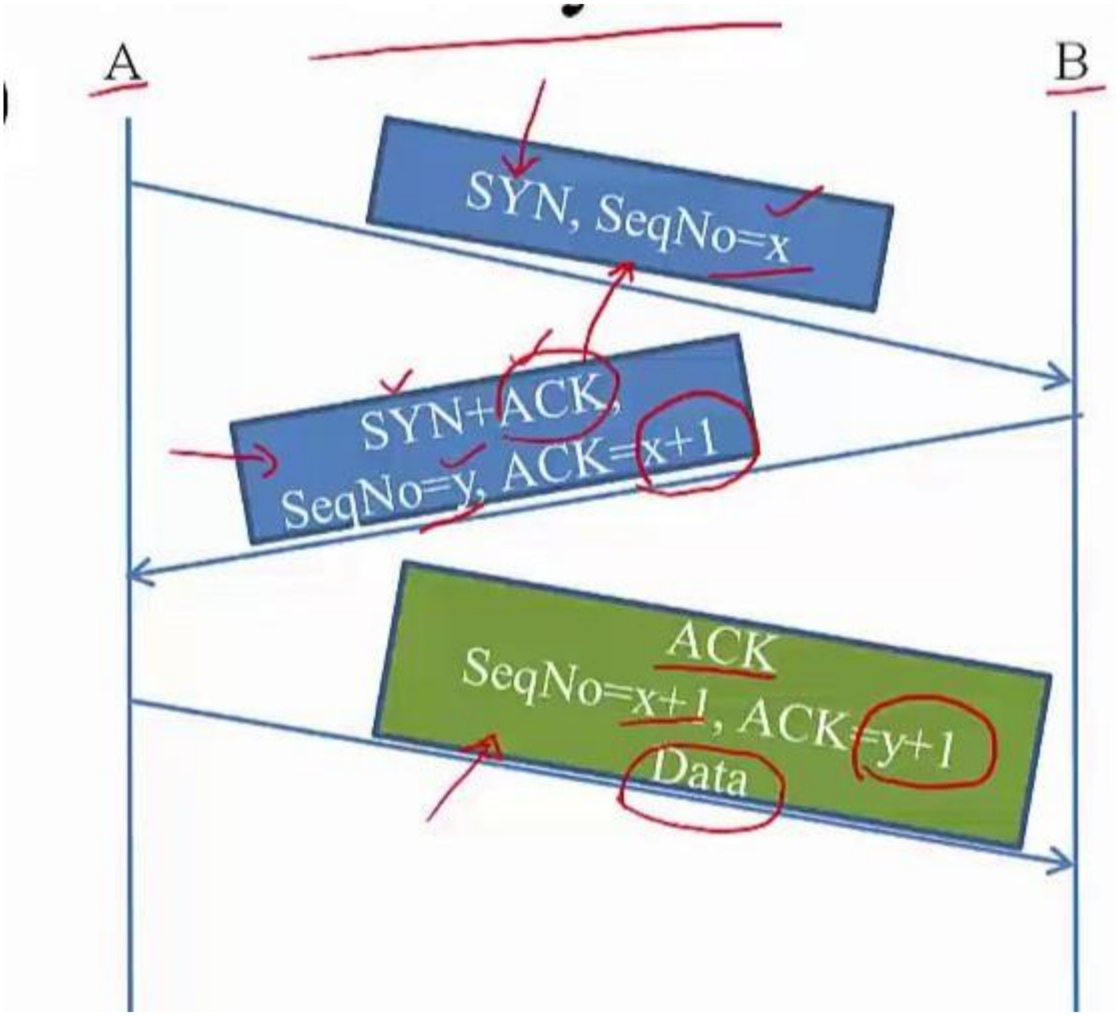
0 – 1023 = System port
1024 – 49151 = User port
49152 – 65535 = Private port

---

### *TCP Connection Management*

- 3 way handshake is used for TCP connection management.
- Client initiates the connection and sends the segment with a sequence number.
- Server acknowledges it back with its own sequence number and ACK of client’s segment.
- Client after receiving ACK of its segment sends an ACK of server’s response.

- Either server or client send TCP segment with FIN flag set to 1.
- The receiving end responds it by acknowledging FIN.
- The TCP communication is closed and connection is released.




---

***Q) For client server application over TCP, server program must be executed before the client program. Why?***

- For TCP applications, as soon as the client is executed, it attempts to initiate a TCP connection with the server. If the TCP server is not running, then the connection can not be established.

---

## Port and Socket

- Port is an endpoint of communication.
- A port is associated with IP address of host and protocol used.
- A port is identified by a 16-bit number known as port number.
Eg: 1.2.3.4:80
IP of host = 1.2.3.4
Port no = 80
Protocol used = TCP

- Socket is one endpoint in a communication flow between two programs running over network.
- It is created and used with a set of programming requests.
- It is also used for communication between processes within the same computer.

---

***Q) Why port number is used?***

- Port number provides a way to identify a specific process to which the message is to be forwarded when it arrives at a server.

---

## Congestion Control Algorithm - Token Bucket and Leaky Bucket

- Congestion is the reduction of quality of service that occurs when a network node is carrying more data than it can handle.

---

***Token Bucket Algorithm:***

- It is an algorithm used in packet switched networks. For congestion control mechanism.
- A bucket is of fixed capacity.
- A token is added to the bucket every 1/r seconds.
- The bucket can hold at most b tokens.
- If a token arrives when the bucket is full, it is discarded.
- When the packet of n bytes arrives, n tokens are removed from the

bucket, and the pocket is sent to the network.
- If fewer than n tokens are available, no tokens are removed from the bucket and the packet is considered to be non-conformant.

---

***Leaky Bucket Algorithm:***

- It is used to determine whether same sequence of events conforms to defined limits.
- A fixed capacity bucket associated with each user, leaks at a fixed rate.
- If bucket is empty, it stops leaking.
- For a packet to conform, it has to be possible to add a specific amount of water to the bucket.
- If the amount of water could cause the bucket to exceed its capacity, the packet does not conform and the water in the bucket is left unchanged.

# 6. Application Layer

## Application Layer

- Application layer is the top most layer in OSI model which is responsible for interacting with user and user applications.
- It is the layer where actual communication is initiated and reflected.
- Irrespective of software used, it is the protocol which is considered at application layer used by that software.

---

## Web - HTTP

### *Web Server*

- Web server is a computer system that processes requests via HTTP protocol, used to distribute information on WWW.
- It stores, processes and delivers web pages to clients.
- The communication between client and server takes place using HTTP.

---

### *Hyper Text Transfer Protocol (HTTP)*

- HTTP is an application protocol for distributed, collaborative and hypermedia information system used for data communication in WWW.
- It acts as a request-response protocol in client-server computing model.
- A request message consists of request line, request header field, empty line and optional message body.
- A response message consists of status line, response header field, empty line and optional message body.

---

### *Example Communication over HTTP*

```
Consider that a http client be web browser which request
s www.egnitenotes.com



Client request:
- A client sends a request in request message as:
      GET / HTTP/1.1  (Request line)
      Host: www.egnitenotes.com (Request header field)
                                (Empty line)



Server response:
      HTTP/1.1 200 OK
      Date: Mon, 23 May 2016 22:38:45 GMT
      Content-Type: text/html; charset = UTF-8
      Content-Encoding: UTF-8
      Content-Length: 138
      Last Modified: Wed, 11 April 2015 11:22:32 GMT
      Server: Apache/1.3.3.7(Unix)
      Etag: “3f80f-1b6-3e1cb03b”
      Accept-Ranges: bytes
      Connection: close



      < html >
      < head >
            < title > Egnite Notes < / title >
      < / head >
      < body >
            Welcome to Egnite Notes.
      < / body >
      < / html >
```

# File Transfer - FTP, PuTTY, WinSCP

### *File Transfer Server*

- A file server is a computer responsible for control storage and management of data files.
- It allows users to share information over a network.

---

### *File Transfer Protocol (FTP)*

- FTP is a standard network protocol used for transfer of files from a server to a client using client-server architecture on a network.

---

### *Communication in FTP*

- FTP may run in active or passive mode, which determines how data connection is established.
- In active mode, client starts listening for incoming data connections from server on port M. It sends FTP command PORT M to inform server on which port it is listening. The server then initiates data channel to the client from its port 20.
- In passive mode, the client uses control connection to send PASV command to the server and recieves server IP address and server port no from the server. The client then uses to open a data connection from an arbitary client port to server IP address and server port no received.

---

### *PuTTY*

- PuTTY is a free and open source file transfer application.
- It supports network protocols like SCP, SSH, Telnet, rlogin.
- It allows local, remote or dynamic port forwarding with SSH.
- It provides user control over SSH encryption key and protocol version.

- It comes bundled with command line SCP client (pscp) and SFTP client (psftp).

---

### *WinSCP (Windows Secure Copy)*

- It is a free and open source SFTP, FTP, WebDAV and SCP client for Microsoft Windows.
- It allows secure file transfer between a local and a remote computer.
- It is based on implementation of SSH protocol from PuTTY and FTP protocol from FileZilla.

---

## E Mail - SMTP, POP3, IMAP

### *Electronic Mail*

- Mail server is the computer system that is responsible to forward mails towards its intended recipient.
- Every email that is sent passes through a series of mail servers before reaching recipient.
- Without the series of mail servers, it would be possible to send emails within same domain only.

---

### *Process of sending email*

- The email client connects to the domain’s SMTP server.
- The email client communicate with SMTP and provides your email address, recipient email address and message body.
- The SMTP server processes the recipient’s email address. If domain of sender and receiver is same, the message is routed directly over to domain’s POP3 or IMAP server. Otherwise, SMTP server communicates with other domain’s server.
- SMTP server communicates with DNS to find recipient’s server and the DNS provides IP address.
- The SMTP server can connect to recipient SMTP server by routing

messages along a series of unrelated SMTP servers.
- The recipient SMTP server scans incoming message and forwards to domain’s POP3 or IMAP server if it recognizes the sender’s domain and username.

---

### *Simple Mail Transfer Protocol (SMTP)*

- It is an email transmit text based protocol which moves the email on and across networks using a process called ‘store and forward’.
- It works with Main Transfer Agent to send communication to the right computer.
- It provides a set of codes that simply communicate email messages between email servers.
- When you send out a message, it is turned into strings of text separated by code words that identify the purpose of each section.
- It provides those codes to servers.
- Email server software helps to understand their meaning.
- As message travels towards destination, it passes through a no of computers.
- Each computer stores it before moving on to next computer in the path.
- It is able to transfer text only.

---

### *Support for images in SMTP*

- Multipurpose Internet Mail Extensions can be used to support image messages in SMTP.
- MIME encodes the non-text content into plain text, which can be transmitted via SMTP.
- It consists of MIME header.
- MIME header includes:
a) MIME-Version = indicates MIME formatted messages
b) Content-Type = indicates media type of message content. For image; image/png
c) Content-Disposition = specify presentation styles of mail messages.

d) Content-Transfer-Encoding = indicates whether or not binary to text encoding scheme is used.
- The non ASCII data uses MIME encoded word syntax.
- The syntax uses string of ASCII characters indicating both original character encoding (charset) and content-transfer-encoding used to map bytes of charset into ASCII characters.

---

### *Post Office Protocol 3 (POP3)*

- It is the standard client/server protocol for receiving emails.
- Email is received and held for the user by the Internet server.
- Periodically, the user check their mail box on the server and download any mail.
- As soon as the user downloaded the mail, POP3 deletes the mail on the server.
- It is a kind of 'store and forward' service.

---

### *Internet Message Access Protocol (IMAP)*

- it is the standard protocol for receiving emails in which the stored messages on the mail server can be viewed and manipulated by the end users as though they are stored locally.
- Users can organize messages into folders on the server.
- It is a kind of remote file server.
- It also supports multiple logins.

---

## DNS Concept

### *Domain Name Server*

- DNS is a protocol that helps and supports protocols used by users and works on client/server model.
- It uses UDP protocol for transport layer communication,
- It uses hierarchical domain based naming scheme.

- A DNS server is configured with fully qualified domain names (FQDN) and email addresses mapped with their respective IP addresses.

---

***Working of DNS***

- When you type www.egnitenotes.com into your browser, the browser sends a query over Internet to find website for the requested URL. The query is forwarded firstly to recursive resolver.
- The recursive resolver takes to root server. It provides DNS information about top level domain such as .com
- The TLD DNS server stores address info for second level domains. When query reaches to TLD server, it provides IP address of required DNS.
- The query is then sent to DNS which knows the IP address of full requested domain and is returned to recursive resolver.
- The recursive resolver provides the IP address to the browser.
- Finally, the browser sends a request to IP address of website to retrieve web contents.

---

## Socket Programming

- It is the process of network programming that provides API and interfaces to handle sockets for communicating using standard mechanisms built into network hardware and OS.

---

***Proxy Caching***

- It is the feature of proxy servers that stores content on the proxy server itself, allowing web services to share those resources to more users.
- It allows a server to act as an intermediary between a user and a provider of web content.

# 7. Introduction to IPV6

## IPv6 Advantages

### *Limitations of IPv4*

- Due to address class allocation practices, public IPv4 addresses are becoming scarce. Because of this, it forces deployment of network address translator to share a public IPv4 address among several private addresses. But, NAT adds complexity and also becomes barrier for applications.
- IPv4 works with flat routing infrastructure in which individual address prefixes were assigned and each prefix became a new route in the routing table.
- IPv4 must be configured either manually or through DHCP.
- It do not have built-in security and rely upon Ipsec for security.
- Due to lack of infrastructure, communication with IPv4 mobile node are inefficient.

---

### *IPv6 as replacement of IPv4*

- IPv6 addresses are 128 bits long, creating a huge amount of address space.
- It uses hierarchical routing infrastructure. It results in relatively few routing entries in the routing table.
- It is automatically configure with the host's IPv6 address.
- It supports for Ipsec protocol headers is required. IPv6 packets are not required to be protected with Authentication header (AH) or Encapsulating Security Payload (ESP).
- It is capable of supporting mobility more efficiently.

---

## Packet Fomats

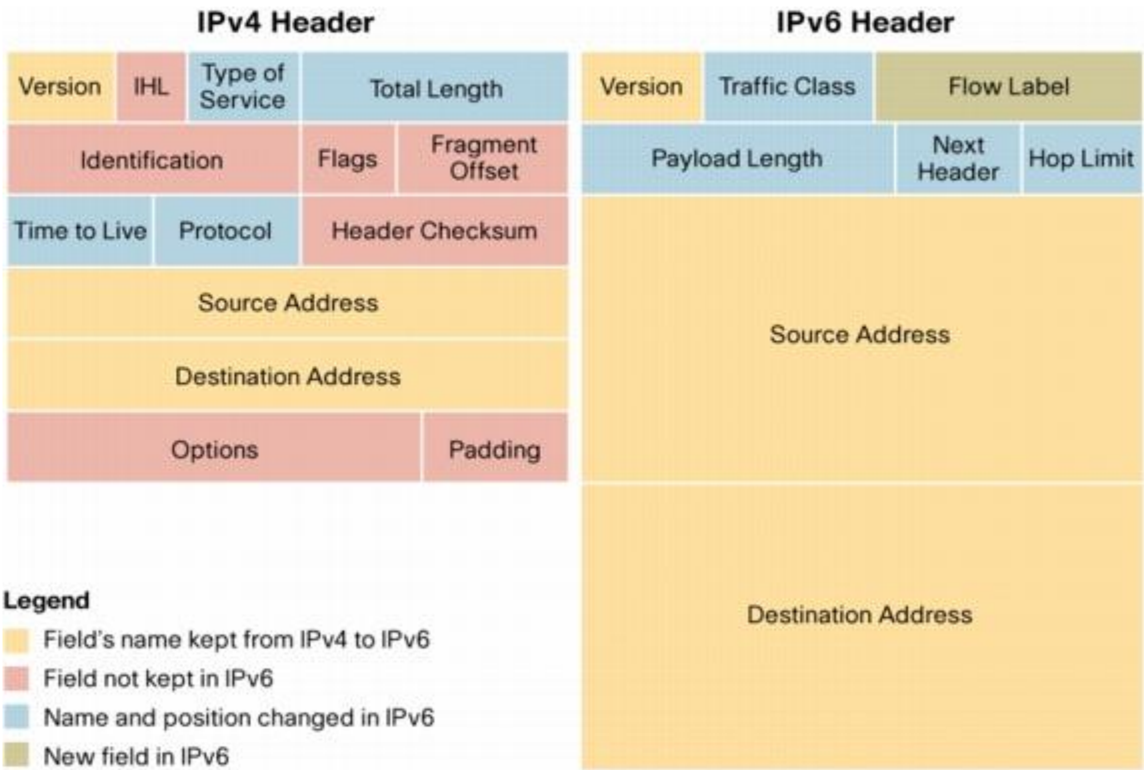
- IP header contains many fields used by routers to forward the packet from network to network to a final destination.
- Payload represents information to be delivered to the receiver by the

sender.
- The basic packet structure is:
| Header | Payload |

---

### *Header Structure*

The header structure of IPv4 and IPv6 are shown in given figure:




---

### *Header Structure Comparison*

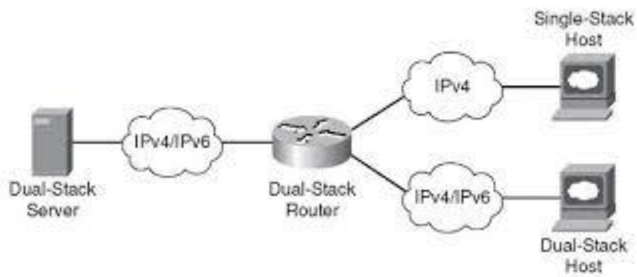
- IPv6 header is simpler than IPv4 header.
- IPv6 header size is bigger than that of IPv4.
- The source and destination addresses are 32 bit in IPv4 header while 128 bit in IPv6 header.
- IPv4 header is of variable size with minimum of 20 byte in length. IPv6 header is of fixed size with 40 byte in length.

---

# Transition from IPv4 to IPv6

### *Dual Stack Router:*

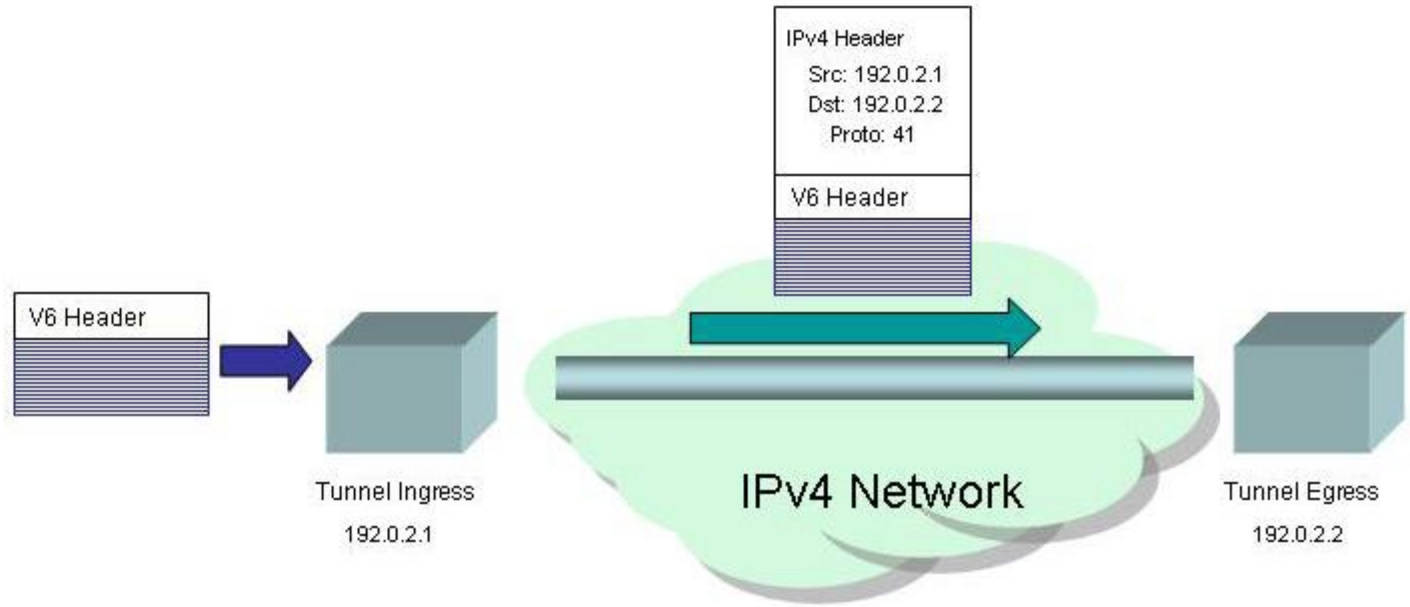
- A router can be installed with both IPv4 and IPv6 address configured on its interfaces pointing to the network of relevant IP scheme.
- It is capable of communicating with both types of network.
- It provides a medium for the hosts to access a server without changing their respective IP versions.
- IPv6 is not supported on all of the IPv4 devices.




---

### *Tunneling:*

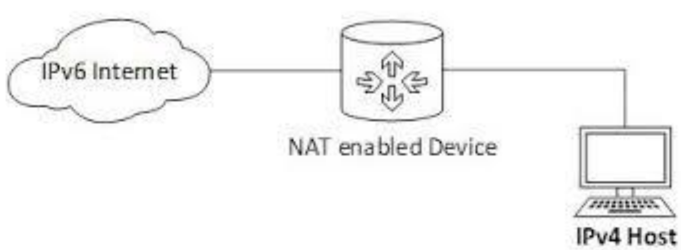
- Tunneling provides a solution where user data can pass through a non supported IP version.
- Two remote IPv4 networks can communicate via a tunnel where the transit network is on IPv6.
- Two remote IPv6 networks can communicate via a tunnel where the transit network is on IPv4.




---

***NAT Protocol Translation:***

- It is the important method of transition to IPv6 by means of a Network Address Translation – Protocol Translation (NAT – PT) enabled device.
- With the help of NAT – PT device, actual translation can take place between IPv4 and IPv6 packets and vice versa.
- When IPv4 host sends a request packet to IPv6 server, NAT – PT device strips down IPv4 packet, removes IPv4 header and adds IPv6 header and passes it through the Internet.
- When a response from IPv6 server comes for the IPv4 host, the router does the reverse action.



***Comparisons:***

- Dual stack allows IPv4 and Ipv6 to coexist in the same devices and networks.
- Tunneling allows transport of IPv6 traffic over the existing IPv4 infrastructure.
- Translation allows IPv6 only nodes to communicate with IPv4 only nodes.

# 8. Network Security

## Network Threats

### *Vulnerabilities and Attacks*

Physical Layer:

- The vulnerability in wired and wireless networks is an unauthorized access to a network through insecure hub/switch port.
- The vulnerability may be exploited to launch attacks like:
a) Sniffing packet data to steal valuable information.
b) Denial of service to legitimate users.
c) Spoofing MAC of legitimate hosts and launch man in the middle attack.

Network Layer

- The vulnerability of TCP/IP is:
a) The plain text data packets exchanged between server and client can be easily read which uses HTTP.
b) HTTP has weak authentication for session initialization which may lead to session hijacking.
c) 3 way handshake for connection establishment is another vulnerability which is exploited by SYN flooding.
d) IP protocol header modification may lead to IP spoofing attack.

---

### *Network Security*

- It aims to make the entire network secure.
- It protects usability, reliability, integrity and safety of network and data.
- The primary goal of network security are:
a) Confidentiality
b) Integrity
c) Availability

---

# Cryptography - Symmetric and Public Key

### *Principles of Cryptography*

- Cryptography is the method of storing and transmitting data in a format that can be rad and processed by the intended users only.
- It includes methods to hide information in storage or in transit.
- It is often associated with encryption and decryption.
- Encryption is the process of scrambling plain text into cipher text.
- Decryption is the process of regaining the plain text from the cipher text.

---

### *Symmetric vs Asymmetric Key Encryption*

- Symmetric key encryption uses a single private key.
- Encryption and decryption both utilizes the same key.
- It does not ensure confidentiality.
- It is generally used for bulk data encryption.
- It is more efficient.
- The algorithms used are AES, DES, Blowfish.

- Asymmetric key encryption uses a pair of keys: private key and public key.
- Encryption is done with public key and the holder of private key can only decrypt it.
- It ensures confidentiality.
- It is generally used to secure key exchanges.
- It is less efficient.
- The algorithm used is RSA.

---

# RSA Algorithm

- RSA is an algorithm used for public key encryption.
- It is believed to be secure if its keys have a length of at least 1024 bits.

***Key Generation Algorithm***

1. Choose two very large random prime integers. (p and q)
2. Compute n and Ǿ(n) such that:
n = p * q
Ǿ(n) = (p-1) * (q-1)
3. Choose an integer e, 1 < e < Ǿ(n), such that:
gcd(e, Ǿ(n)) = 1
4. Compute d, 1 < d < Ǿ(n), such that:
e * d = 1 (mod Ǿ(n))

We get, public key = (n, e)
private key = (n, d)
p, q and Ǿ(n) are private
e is public exponent.
d is private exponent.

---

***Encryption and Decryption:***

- Cipher ( C ) = M^e mod n
- Message ( M ) = C^d mod n

---

***Example:***

```
Key Generation:
1. p = 11 and q = 3
2. n = p * q = 33
    Ǿ(n) = (p-1) * (q-1) = 20
3. Choose e = 3 such that 1 < e < Ǿ(n) and gcd(e, Ǿ(n))
= 1
4. (3) * d = 1 (mod 20)
    d = 7
```

```
Public key = (33, 3)
Private key = (33, 7)


Let message M = 7.


Encryption:
C = M^e mod n = 7^3 mod 33 = 13


Decryption:
M = C^d mod n = 13^7 mod 33 = 7
```

---

## Digital Signature

- Digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software or digital document.
- It is intended to solve the problem of tampering and impersonation in digital communication.
- It is based on public key cryptograhy.
- Using RSA, public and private keys are generated.
- The signing software is used to create a one-way hash of the electronic data to be signed.
- The private key is used to encrypt the hash.
- The encrypted hash along with hashing algorithm is the digital signature.

---

## Securing E-mail : PGP

***Pretty Good Privacy (PGP)***

- It is a program used to encrypt and decrypt email over the Internet as well as authenticate messages with digital signatures.
- Each user has encryption key and private key.
- Message is encrypted and send to someone using their encryption key.
- It uses faster encryption algorithm to encrypt message.
- The receiver private key is used to decrypt to short key; which is the key used to decrypt the message.

---

## Securing TCP Connections - SSL

- SSL is a standard security technology for establishing encrypted link between a server and a client.
- It allows sensitive informations like credit card no, social security no, etc to be transmitted securely.
- Generally, the data between browser and web server is sent in plain text, which is vulnerable if the intruder intercept the data.
- It provides variables of the encryption for both the link and the data being transmitted.
- SSL secured websites begin with https.
- SSL certificates have public and private key, which work together to establish an encrypted connection.
- When a browser attempts to access a SSL secured websites, the browser and the web server establish an SSL connection using SSL handshake.
- Public key, private key and session key are used to set up SSL connection.
- After secure connection is made, session key is used to encrypt all transmitted data.

---

## Network Layer Security - IPsec and VPN

***Internet Protocol Security ( IPSec )***

- IPSec is a framework for a set of protocols for security at the network or packet processing layer of network communication.
- It is useful for implementing VPN and for remote user access through dial up connection to private networks.
- The advantage is that, security arrangements can be handled without requiring changes to individual user computers.
- It provides two security service:
a) Authentication header (AH) = allow authentication of data sender.
b) Encapsulating Security Payload (ESP) = allow sender authentication as well as data encryption.

---

***Virtual Private Network (VPN)***

- It is a technology that creates safe and encrypted connection over a less secure network.
- It allows remote users to securely access applications and other resources.
- Data travels through secure tunnel.
- VPN users must use authentication method to gain VPN access.
- It ensures appropriate level of security to connected system.
- Speed of Internet connection of user effects VPN performance.

---

## Securing Wireless LAN - WEP

- It is a security protocol specified by IEEE WiFi standard, which provides a wireless LAN with a level of security and privacy comparable to what is expected on a wired LAN.
- Data encryption protects vulnerable wireless link between clients and access points.
- After this measure, password protection, end to end encryption and VPN can be placed to ensure privacy.
- It is vulnerable to wireless equivalent privacy attacks.

---

# Firewalls

- Firewall is a network security that analyzes and controls the incoming and outgoing network traffics based on the predetermined security rules.
- It analyzes the network and allows the network traffics in and out only if they are trusted.

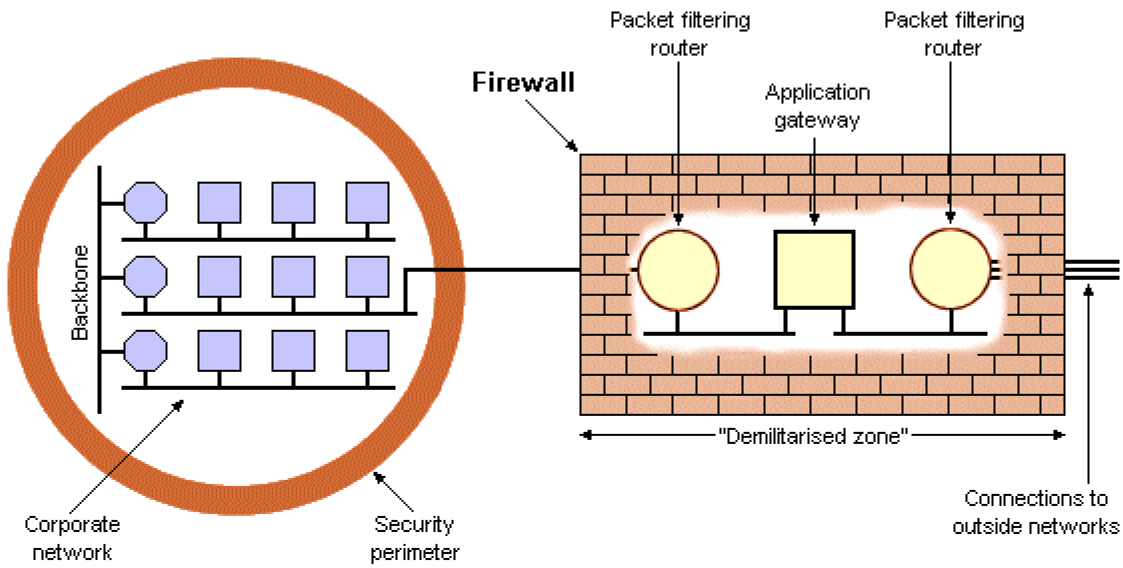
---

***Types of Firewall***

1) Packet Filtering Firewall
- It protects users from the external network threat
- Packet filtering is the process of passing or blocking packets based on source and destination address, port or protocols at a network interface.
- The header of the packet is analyzed and based on predefined rules, it allows packet to pass or prevents packet from passing.

Methods
1) The filter accepts only those packets that it is certain are safe, dropping all others.
2) The filter drops only the packets that is certain are unsafe, accepting all others.
3) The filter when encounters a packet for which no rule is provided, it query the user for performing what should be done.

2) Application Gateway
- The gateway operates at the application layer.
- Application gateway for specific applications can be installed.
- It filters incoming node traffic to certain rules which mean that only transmitted network application data is filtered.
- Eg: A mail gateway can be set up to examine each message going in or coming out. For each message, gateway decides whether to transmit of discard the messages.


Packet filtering router
Packet filtering router
Firewall
Application gateway
Backbone
"Demilitarised zone"
Corporate network
Security perimeter
Connections to outside networks